Knowledge check · Pre

- **Not graded**, and anonymous
- On your quiz sheet or a piece of paper, write your **code word** and **Pre**
- For each question, write its number and your answer (e.g. “1 b”)
- One question at a time on the screen

KEY MESSAGE

It’s fine not to know an answer. Just leave it blank.

KNOWLEDGE CHECK · QUESTION 1 OF 15

Which question can whole-genome sequencing answer that standard culture + AST cannot?

a) Which antibiotic should this patient receive today

b) Whether isolates from several patients are the same strain

c) Whether the organism is Gram-positive

d) Whether the patient has a fever

KNOWLEDGE CHECK · QUESTION 2 OF 15

Do you need to own a sequencer to do genomic surveillance? (Yes / No) Briefly explain why.

Write your answer on your paper

KNOWLEDGE CHECK · QUESTION 3 OF 15

Name two bacterial species that commonly cause surgical site infections.

Write your answer on your paper

KNOWLEDGE CHECK · QUESTION 4 OF 15

An Illumina sequencing run typically produces reads that are:

a) 100–300 bases long and highly accurate

b) 10,000+ bases long

c) the whole genome in one read

d) protein sequences

KNOWLEDGE CHECK · QUESTION 5 OF 15

Which technology is better at showing whether a resistance gene is on a plasmid?

a) Illumina short reads alone

b) Oxford Nanopore long reads

c) AST

d) MLST

KNOWLEDGE CHECK · QUESTION 6 OF 15

A genome is 5 million bases. You have 1 million reads of 150 bases. What is the approximate coverage?

a) 3×

b) 30×

c) 150×

d) 300×

What does a FASTQ file contain that a FASTA file does not?

a) Sequence names

b) DNA letters

c) Quality scores

d) Gene names

KNOWLEDGE CHECK · QUESTION 8 OF 15

After assembly, which result usually means a *better* assembly?

a) More contigs and a smaller N50

b) Fewer contigs and a larger N50

c) A total length twice the expected genome size

d) No contigs

The gene *mecA* in *S. aureus* indicates:

a) Vancomycin resistance

b) MRSA (methicillin resistance)

c) Carbapenem resistance

d) A virulence toxin

blaNDM-1 is an example of:

a) An ESBL

b) A carbapenemase

c) An efflux pump

d) A housekeeping gene

KNOWLEDGE CHECK · QUESTION 11 OF 15

The genome shows no known resistance gene, but AST says “resistant”. Give one possible explanation.

Write your answer on your paper

Two *Klebsiella* isolates are ST15 and ST307. Could they be part of the same recent transmission chain?

a) Yes, very likely

b) No, they are different strain types

c) Only if they are both ESBL

On a phylogenetic tree, what tells you how genetically different two isolates are?

a) Their order from top to bottom

b) The branch lengths between them

c) The colour of the labels

KNOWLEDGE CHECK · QUESTION 14 OF 15

Four isolates are 2–6 SNPs apart and the patients overlapped on the same ward. This most strongly suggests:

a) Unrelated infections

b) Possible transmission / outbreak

c) Contamination of the sequencer

KNOWLEDGE CHECK · QUESTION 15 OF 15

List three pieces of metadata that must be recorded with every isolate for surveillance.

Write your answer on your paper

KNOWLEDGE CHECK

How confident are you?

Rate each from **1** (not at all) to **5** (very confident):

- I can explain what a genome sequence can tell us about an SSI
- I can run a tool in Galaxy
- I can identify resistance genes from a genome
- I can interpret a phylogenetic tree
- I can plan a genomic surveillance project

N1 · About you

Your main role:
Clinician · Nurse / midwife · Laboratory · Public health / IPC · Research · Other

Your institution:
UGHE · Butaro Hospital · Other

N2 · At your site, is there…

Answer **Yes**, **No** or **Not sure**:

1. culture and antibiotic susceptibility testing?
2. long-term storage of bacterial isolates (e.g. a freezer)?
3. internet reliable enough to use web tools like Galaxy?

N3

What is the biggest barrier to using genomic data in your work?